

Our next goal is to discuss how objects *move* in space. When discussing the movement of objects, the relevant data is the direction you are going and how far you travel in that direction. Mathematically, these geometric objects are called **vectors** and they are *location independent*:

Given $P(x_1, y_1)$ and $Q(x_2, y_2)$, the vector $\mathbf{v} = \overrightarrow{PQ}$ is in *standard position* if its initial point is at the origin. This makes it easy to determine what x and y changes need to occur in order to get to the terminal point. These changes are called the **components** of the vector $\mathbf{v}$.

After shifting $\mathbf{v} = \overrightarrow{PQ}$ to have its initial point at the origin, we denote the subsequent terminal point by (v_1, v_2) . These are exactly the components of $\mathbf{v}$. Thus, the **component form** of $\mathbf{v}$ is

$$\mathbf{v} = \langle v_1, v_2 \rangle = \langle x_2 - x_1, y_2 - y_1 \rangle.$$

In 3-space, given the point $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$, the vector $\mathbf{v} = \overrightarrow{PQ}$ is given by

$$\mathbf{v} = \langle x_2 - x_1, y_2 - y_1, z_2 - z_1 \rangle.$$

Fact: Since vectors are location independent, two vectors $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$ and $\mathbf{u} = \langle u_1, u_2, u_3 \rangle$ are equal if and only if $v_1 = u_1, v_2 = u_2$, and $v_3 = u_3$.

The **magnitude, norm, or length** of a vector $\mathbf{v} = \overrightarrow{PQ} = \langle v_1, v_2, v_3 \rangle$ is given by

$$|\mathbf{v}| = \|\mathbf{v}\| = d(P, Q) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2} = \sqrt{v_1^2 + v_2^2 + v_3^2},$$

where we take the positive square root.

Because of this definition, it becomes clear that the only vector with length 0 is the zero vector $\mathbf{0} = \vec{0} = \langle 0, 0, 0 \rangle$. Additionally, this is the only vector with no specific direction.

Example. Consider the points $P(-3, 4, 1)$ and $Q(-5, 2, 2)$. Form the vector $\mathbf{v} = \overrightarrow{PQ}$.

(i) Find the component form of $\mathbf{v}$.

(ii) Find the magnitude of $\mathbf{v}$.

Definition. Let $\mathbf{u} = \langle u_1, u_2, u_3 \rangle$ and $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$ be vectors and let k be a scalar (i.e., a real number). Then,

$$\mathbf{u} + \mathbf{v} = \langle u_1 + v_1, u_2 + v_2, u_3 + v_3 \rangle \quad \text{and} \quad k\mathbf{u} = \langle ku_1, ku_2, ku_3 \rangle.$$

Example. With $\mathbf{u} = \langle 1, 2 \rangle$ and $\mathbf{v} = \langle 3, 1 \rangle$, compute $\mathbf{u} + \mathbf{v}$ and $-2\mathbf{u}$, and draw relevant pictures.

Oftentimes, when adding two vectors, we can visually see their sum by moving the vectors ‘head-to-tail’ and then taking the straightest path to the end. This is sometimes called the **parallelogram law** as well. The sum is oftentimes called the *resultant vector*.

Example. Let $\mathbf{u} = \langle -1, 3, 1 \rangle$ and $\mathbf{v} = \langle 4, 7, 0 \rangle$. Find the components and norm of $2\mathbf{u} - 3\mathbf{v}$.

There are other ways to represent vectors. One route uses **unit vectors** (vectors with norm 1). In 3-space, there are some obvious ones:

These are called the **standard unit vectors**. Generally speaking, any vector $\mathbf{v}$ can be rewritten as

$$\mathbf{v} = \langle v_1, v_2, v_3 \rangle = v_1 \langle 1, 0, 0 \rangle + v_2 \langle 0, 1, 0 \rangle + v_3 \langle 0, 0, 1 \rangle = v_1 \mathbf{i} + v_2 \mathbf{j} + v_3 \mathbf{k},$$

where v_1 , v_2 , and v_3 are the **i**-component, **j**-component, and **k**-component, respectively.

There is yet another way to represent a vector. Namely, given an arbitrary non-zero vector $\mathbf{v}$, if we want to know its directional unit vector, we simply need to compress it by a factor of its length, $|\mathbf{v}|$. Specifically, the unit vector

$$\frac{\mathbf{v}}{|\mathbf{v}|}$$

points in the direction of $\mathbf{v}$. In this manner, **every** vector $\mathbf{v}$ can be written as a product of its length and its direction:

$$\mathbf{v} = |\mathbf{v}| \frac{\mathbf{v}}{|\mathbf{v}|}.$$

Example. If $\mathbf{v} = \langle 3, -4 \rangle = 3\mathbf{i} - 4\mathbf{j}$ is a velocity vector, express $\mathbf{v}$ as a product of its speed times its direction of motion.

Example. Find a vector of magnitude 3 in the direction of $\mathbf{v} = 10\mathbf{i} - 24\mathbf{k}$.